Layer 1

Global DNS Routing

Where on the internet does play.petflix.com live?

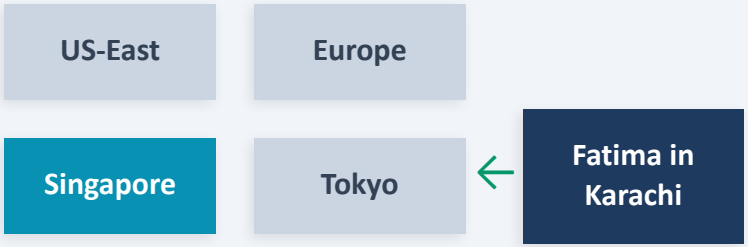
DNS: Routing Fatima to Singapore

"What is the IP address of play.petflix.com?"

Petflix DNS Intelligence

- Sees resolver IP (Karachi)
- Checks data center health & load
- Returns nearest healthy region IP
- **Karachi → Singapore (closest)**

Regional Routing



DNS Failover & Load Balancing

If Singapore Goes Down

- Health checks detect failure in seconds
- DNS stops returning Singapore's IP
- Next lookup routes to Hong Kong or Tokyo

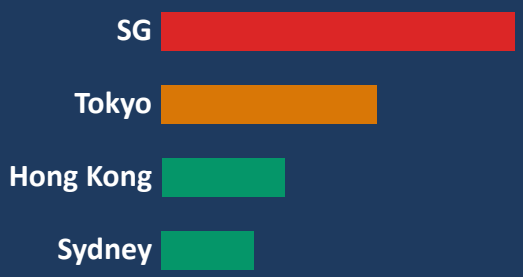
SG ✗ → Hong Kong

The TTL Problem

- DNS answers are cached by resolvers
- Cached IP may point to a dead region
- **Solution: keep TTLs short (30s)**
- Trade-off: more DNS traffic for faster failover

Geographic Bottleneck: Singapore

Singapore absorbs traffic from all of South and Southeast Asia. Smart DNS sometimes routes users to a slightly farther region to avoid overloading a hot data center.



Layer 2

API Gateway

PetGate: the front door to all of Petflix's backend

PetGate: Routing & Service Discovery

Fatima's Request → Load Balancer → PetGate API Gateway → Service Registry

Service Registry
- Every instance registers on startup
- Periodic heartbeats

1. Authenticate • 2. Route to Service • 3. Pick Healthy Instance

PetGate routes to healthy playback service instances

Playback #1	Playback #2	Playback #3	Playback #4
✓ Healthy	✓ Healthy	✗ Slow (5s)	✓ Healthy

Circuit Breaker: traffic to slow instance #3 is cut off

Circuit Breaker Pattern

CLOSED

Traffic flows normally.
PetGate monitors error rates.

→ *>50% errors*

OPEN

Traffic stopped.
Fallback response returned.
Cooldown period.

→ *cooldown expires*

HALF-OPEN

Single test request sent.
Success → close circuit.
Failure → stay open.

✓ success → CLOSED ✗ failure → stays OPEN

Bulkhead Principle: isolate failures so they don't spread. A slow playback instance shouldn't take down recommendations.